Integrating VirusTotal with Wazuh SIEM for Threat Intelligence

By Bensekhar Maria Golsaf

Prerequisites

Before starting, ensure you have the following:

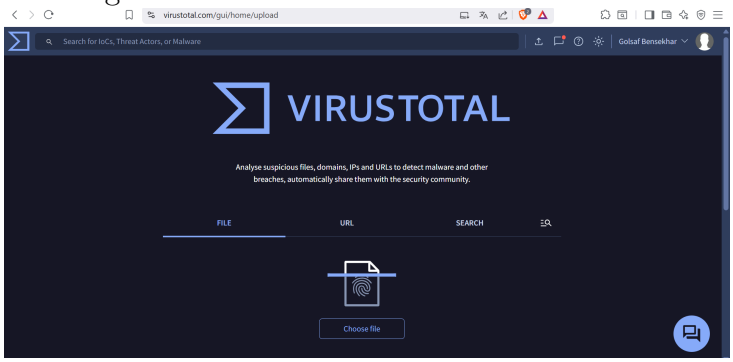
1. A running **Wazuh SIEM** setup
2. At least one **Wazuh agent** connected to the Wazuh manager
3. A valid **VirusTotal API key**

Introduction

In a modern Security Operations Center (SOC), analysts face hundreds of alerts daily. Without proper context, it can be difficult to distinguish between false positives and real threats. This is where **threat intelligence** becomes essential, it enriches raw alerts with external data, helping analysts prioritize incidents more effectively.

VirusTotal is a widely used threat intelligence platform that analyzes files, URLs, domains, and IPs against multiple antivirus engines. By integrating VirusTotal with Wazuh, you can automatically enrich alerts with reputation data, making investigations faster and more accurate.

Note: Any file or URL uploaded to VirusTotal should be considered public. Do not upload sensitive or proprietary files, as they may be accessible to other organizations.



Understanding VirusTotal

VirusTotal allows security professionals to:

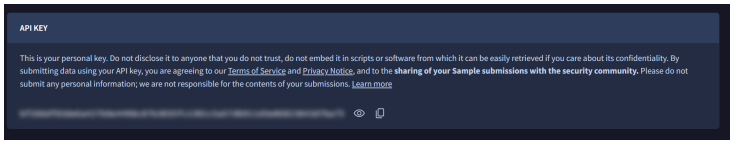
- **Upload and scan files or URLs** using dozens of antivirus engines
- **Retrieve file and URL hashes** for malware tracking

- **Extract metadata** to aid in investigations
- **Leverage the VirusTotal API** to automate lookups and integrate with SIEM platforms like Wazuh

This integration enables Wazuh to automatically query VirusTotal when new files are detected on endpoints, enriching alerts with reputation scores.

Step 1: Obtain a VirusTotal API Key

1. Create an account on VirusTotal.
2. Navigate to **Profile** → **API Key**.
3. Copy your personal API key for later use.

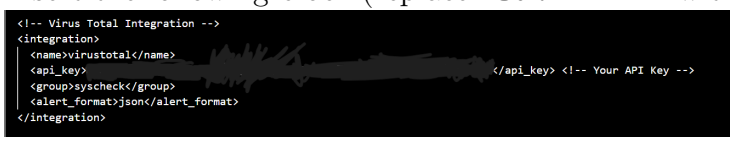


Step 2: Configure Wazuh Manager

On your **Wazuh Manager**, edit the main configuration file:

```
Open C:\Program Files (x86)\ossec-agent\ossec.conf
with a text editor (e.g., Notepad) and add the configuration
    snippet.
```

Insert the following block (replace `YOUR_API_KEY` with your actual key):




```
<integration>
  <name>virustotal</name>
  <api_key>YOUR_API_KEY</api_key>
  <alert_format>json</alert_format>
</integration>
```

Save the file, then restart the Wazuh Manager:

```
net stop wazuhsvc
net start wazuhsvc
```

Step 3: Configure the Wazuh Agent

On the **Wazuh Agent**, define which directory should be monitored in real-time. Edit the configuration file:

```
Open C:\Program Files (x86)\ossec-agent\ossec.conf
with a text editor (e.g., Notepad) and add the configuration
   snippet.
```

Add the following snippet:

```
<directories check_all="yes" realtime="yes">/home/virustotaltest</directories>
```

```
<syscheck>
  <directories realtime="yes">/home/virustotaltest</
     directories>
</syscheck>
```

This ensures that any file added to `/home/virustotaltest` will trigger a VirusTotal lookup.

Restart the agent to apply changes:

```
net stop wazuhsvc
net start wazuhsvc
```

Conclusion

By integrating **VirusTotal with Wazuh**, you extend your SOC's detection and analysis capabilities with external threat intelligence. This setup enables:

- Automatic reputation lookups for files detected by Wazuh agents
- Faster triage of alerts by distinguishing benign from malicious files
- Enhanced visibility into potential threats with minimal manual effort

This integration is ideal for SOC analysts, incident responders, and cybersecurity students who want hands-on experience with **threat intelligence enrichment in SIEM workflows**.